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SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

***CO1:** Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*
Assessment Tool Weightage: 40%

Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

1 hr = 60 min
10 x 5 = 50

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every

window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing
Understanding of Concepts	45%	2.25 Demonstrates a thorough, accurate understanding of concepts	2 Good understanding with minor gaps	1 Basic understanding with some misconceptions
Explanation	35%	1.75 Detailed, well-explained answers with relevant examples	1 Clear explanation with adequate details	0.5 Limited explanation, lacks depth.
Clarity	20%	1 Clear, neat, professional presentation	0.5 Understandable with minor issues	0 Limited clarity, readability issues

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	45% ~~25%~~	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps 2	Basic understanding with some misconceptions 1	Poor understanding; major misconceptions
Explanation	35% ~~25%~~	Detailed, well-explained answers with relevant examples 1.75	Clear explanation with adequate details 1	Limited explanation; lacks depth 0.5	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	20% ~~5%~~	Clear, neat, professional presentation 1	Understandable with minor issues 0.5	Limited clarity, readability issues 0	Poorly written, difficult to understand

5

	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
	23/2/21	23/2/21	23/2/21	23/2/21	23/2/21	23/2/21	23/2/21	23/2/21	23/2/21	23/2/21
Understanding 45%	2.25	2	2.25	2.	2.25	2	2	1	2	2.25
Explanation 35%	1	1	1.75	1.75	1.75	1	1	0.5	1.75	1
Clarity 20%	1	0.5	0.5	1	1	0.5	0.5	1	0.5	0.5
	4.25	3.5	4.5	3.75	5	3.5	3.5	2.5	4.25	3.75

$\frac{38.5}{50}$

77%

CO1 - Assessment Tool - 1

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Course : Computer Networks.

① Soln :

Gin :

* File size = 150MB = 150×10^6 = 150,000,000 bytes
* Segment size = 1500 bytes
* Data Link Layer overhead = 40 bytes / frame
* Bandwidth = 50 Mbps.

1. No of segments

$$\frac{150{,}000{,}000}{1500} = 100{,}000 \text{ segments}$$

Each segment is placed in one frame.

Frames transmitted = 100,000

2. Total data transmitted

150MB + 4MB = 154MB

In bits :

$$154 \times 10^6 \times 8 = 1232 \times 10^6 \text{ bits}$$

3. Total Data Link Layer overhead :

100,000 × 40 = 4,000,000 bytes = 4MB

4. Transmission time :

$$\frac{1232 \times 10^6}{50 \times 10^6} = 24.64 \text{ seconds}$$

Reliability

* Transport Layer : Ensures reliable delivery using segmentation, sequence no's, acknowledgements & flow control
* Data Link Layer : Ensures error-free communication over a single link using framing, error detection & retransmission of corrupted frames.

Flow chart:

150MB Design File

↓

Transport Layer
Divides file into 1500-byte segments

↓

100,000 segments created

↓

Data Link Layer
Adds 40-byte Header + Trailer to each

↓

100,000 Frames Transmitted
(1500 bytes + 40 bytes = 1540 bytes / frame)

↓

Sent over Network
Bandwidth = 50 Mbps.

2. Sliding Window Protocol

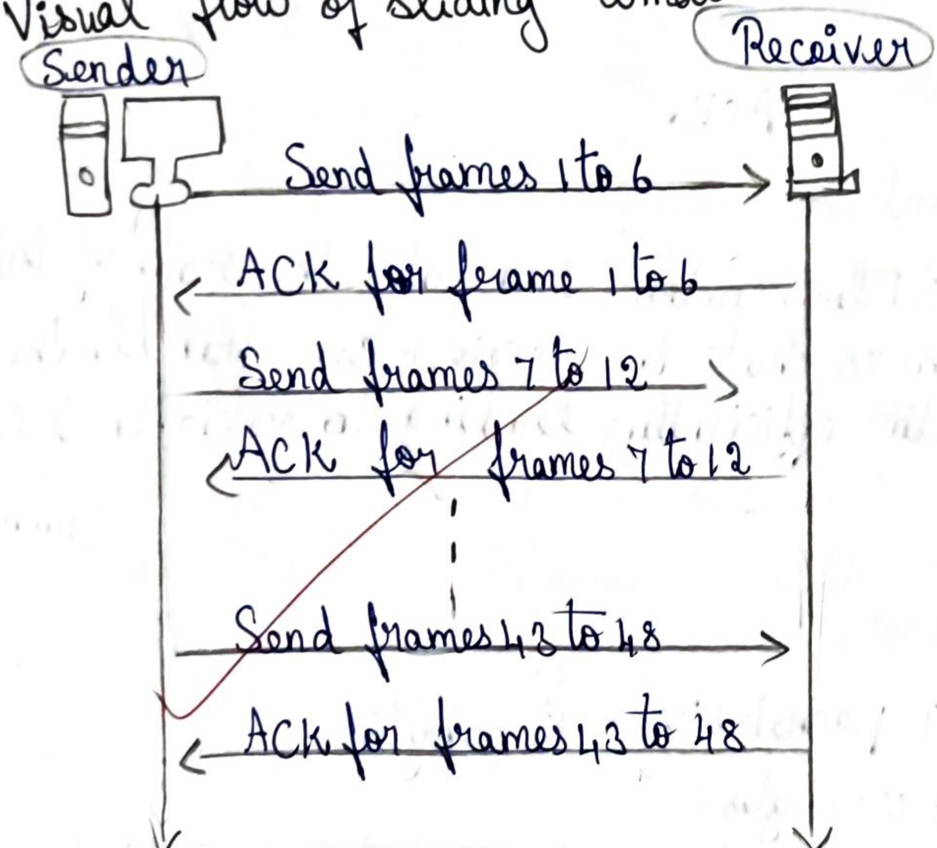
Gvn:

* Window size = 6 frames
* Data per frame = 1024 bytes
* Total frames = 48 frames
* 1 frame is lost in every window

What is Sliding Window ?

The sender can send up to 'window size' number of frames without waiting for ACK. After recieving ACKs, the window slides forward.

Visual flow of sliding window.



If any frame is lost, sender re-transmits that frame after timeout or on receiving -ve ACK

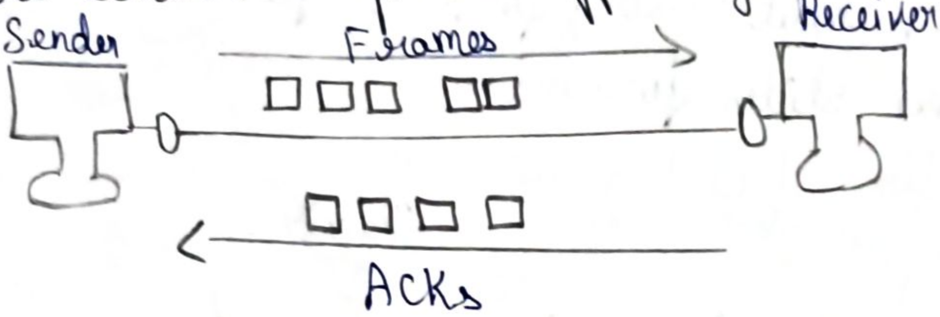
Calculations:

1. No of transmisson windows = Total frames / window size = $48/6$ = 8 windows

2. Lost frames = 1 frame in each window = 8 frames
3. Total retransmissons = No of windows × Lost frames = 8 × 1 = 8 retransmiss

4. User data pere frame = 1024 bytes (given)
5. Total user data sent = 48 × 1024 = 49,152 bytes
6. Total data including retransmissions = (48+8) × 1024
= 57,344 bytes.

Flow Control - Improves efficiency.



How Flow Controls Help?

↳ Flow control regulates the rate of data transmission so that the receiver can handle the incoming data efficiently, leading to reliable & efficient Communication

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③ Gin:

* Total IP packet size = 12000 bytes
* MTU = 1500 bytes
* IP header per fragment = 20 bytes.

Step 1: Maximum data per fragment
[MTU includes header + data] → MTU - IP header
= 1500 - 20 = 1480 bytes

Step 2: No of fragments

Total data to be sent = Original Packet size - Original IP header.

$= 12000 - 20 = 11980$ bytes

No of fragments $= \dfrac{\text{Total data}}{\text{Max data / fragment}} = \dfrac{11980}{1480} = 8.094$

Per

Total no of fragment = 9 (as we cannot have fraction of fragments).

Step 3: Total header overhead → No of fragment × Header size
[Each fragment has 20-bytes header]

$= 9 \times 20 = 180$ bytes.

How the Networ Layer Ensures Successful Delivery.

The Network Layer (IP Layer) ensures successful delivery of fragmented packets through the following mechanisms:

* Fragment Identification: Identifies all fragments as belonging to the same original packet.
* Fragment Offset: Indicates the correct position of each fragment in the original packet.
* More Fragments (MF) Flag: Shows whether more fragments follow or if it is the last fragment.
* Reassembly: The destination combines all fragments to reconstruct the original packet.
* Timeout: If all fragments do not arrive within the time limit, the incomplete packet is discarded.

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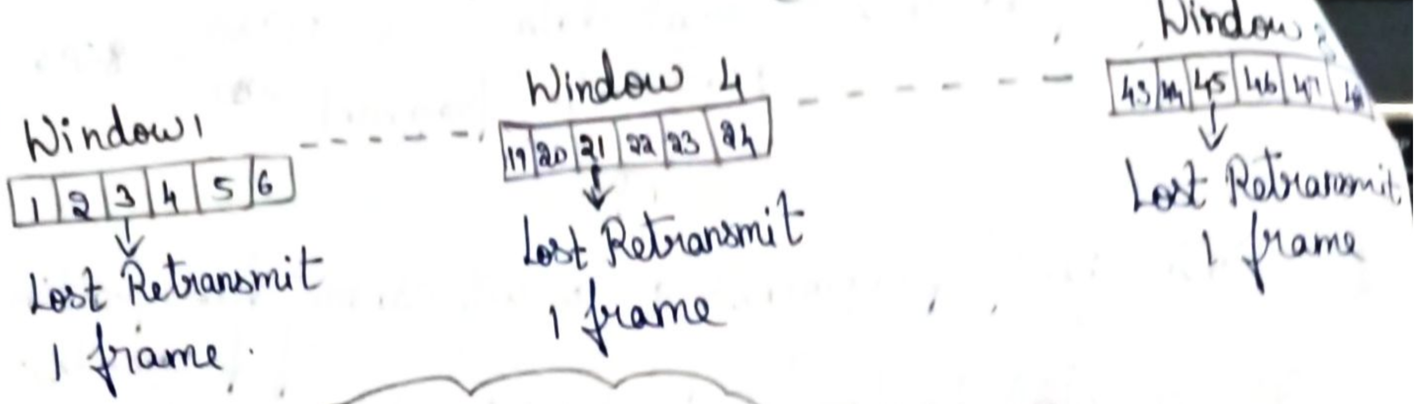
(4) Gn:

* Window size = 6 frames
* Data per frame = 1024 bytes
* Total frames = 48
* 1 frame is lost in every window.

Sliding Window Transmission
(Window size = 6 frames)



Total Windows = 8

Calculations:

1. No of transmission Windows = Total frames / Window size = 48/6 = 8

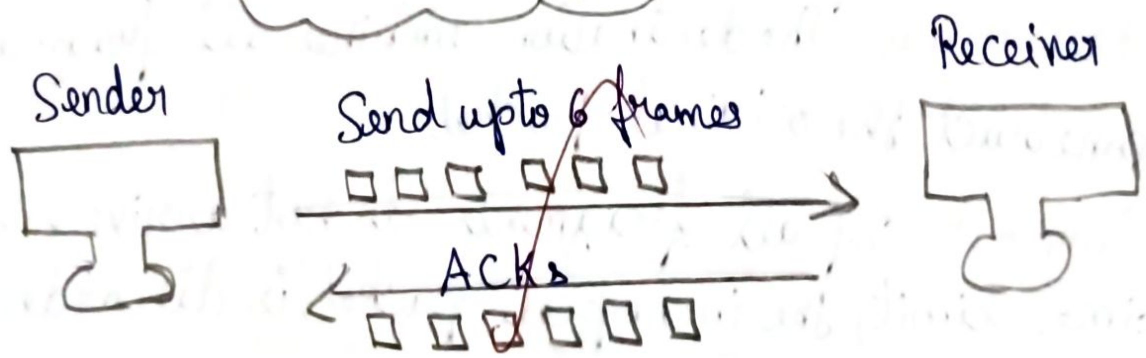
   = 8 windows

2. Total No of retransmissions = 1 frame lost in every window

   = 1×8 = 8 retransmissions

3. Total user data sent = Total frames × Data/frame = 48×1024

   = 49,152 bytes



- ★ It prevents receiver buffer overflow
- ★ It allow the receiver to process data at its own pace.
- ★ It reduces congestion & packet loss.

Gin :

Total file size = 600MB

Checkpoint inserted after every = 60 MB

Failure occurs after = 390 MB

* No of checkpoints created before failure :

$= 390/60 = 6.5$

Full checkpoints only created, so it is 6

* Amount of data to be retransmitted after recovery :

Last successful checkpoint $= 6 \times 60 = 360$ MB

Data after last checkpoint $= 390 - 360 = 30$ MB.

So, data to retransmitted $= 30$ MB

Importance of synchronization in the Session Layer :

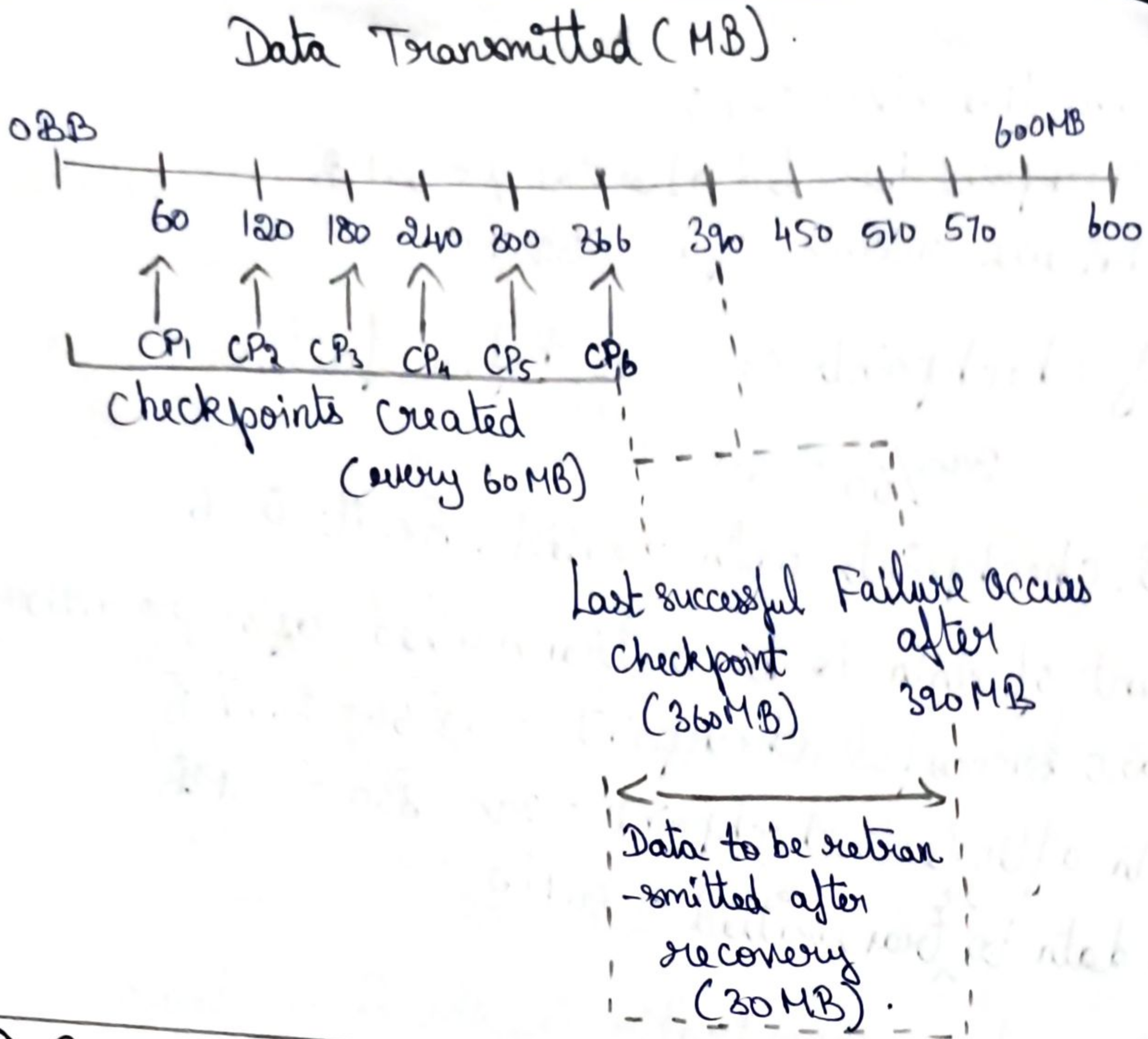
* Recovery Efficiency : Allow the system to restart transmission from the last checkpoint instead of the beginning, saving time & bandwidth.

* Data Consistency : Ensures data is delivered in the correct order & without loss.

* Error Handling : Helps detect and recover from failures during transmission.

* Resource Management : Prevents unnecessary retransmission of already successfully received data

* Smooth Communication : Maintains continuous & reliable session between sender & receiver.



(6) Gin :

Original file size = 900 MB

Compression = 40%

Encryption overhead = 5% of the compressed size

<u>Calculations</u>

1. Compressed File Size

$\Rightarrow 900 \times (1 - 0.40) = 900 \times 0.60 = 540\,MB$.

2. Encrypted File Size

Encryption Overhead $= 540 \times 0.05 = 27\,MB$

Encrypted File Size $= 540 + 27 = 567\,MB$

3. Total Percentage Reduction $\rightarrow 900 - 567 = 333\,MB$

Percentage Reduction $= \frac{333}{900} \times 100 = 37\%$.

...ow the Presentation layer Improves Transmission Efficiency & Security

The Presentation Layer is responsible for data formatting, Compression, encryption & decryption before transmission

Compression -> reduces the file size, which decreases bandwidth usage & transmission time.

Encryption -> converts the compressed data into a secure format, protecting it from unauthorized access during transmission.

⑦ Gm:

* Total data to be transferred = 3 GB.
* Throughput = 120 Mbps
* Application Layer request size = 3 MB.

Conversions:

* 1 GB = 1024 MB
* 1 MB = 8 Megabits (Mb)
* Throughput = 120 Mbps = 120 Megabits/second.

1. Total Number of Requests.

Total data = 3 GB = 3×1024 MB = 3072 MB.

Request size = 3 MB

$$\text{Number of Requests} = \frac{\text{Total Data (in MB)}}{\text{Request Size (in MB)}}$$

$$= \frac{3072}{3} = 1024 \text{ requests.}$$

2. Estimate Total Transmission Time.

Throughput = 120 Mbps = 120×10^6 bits/second

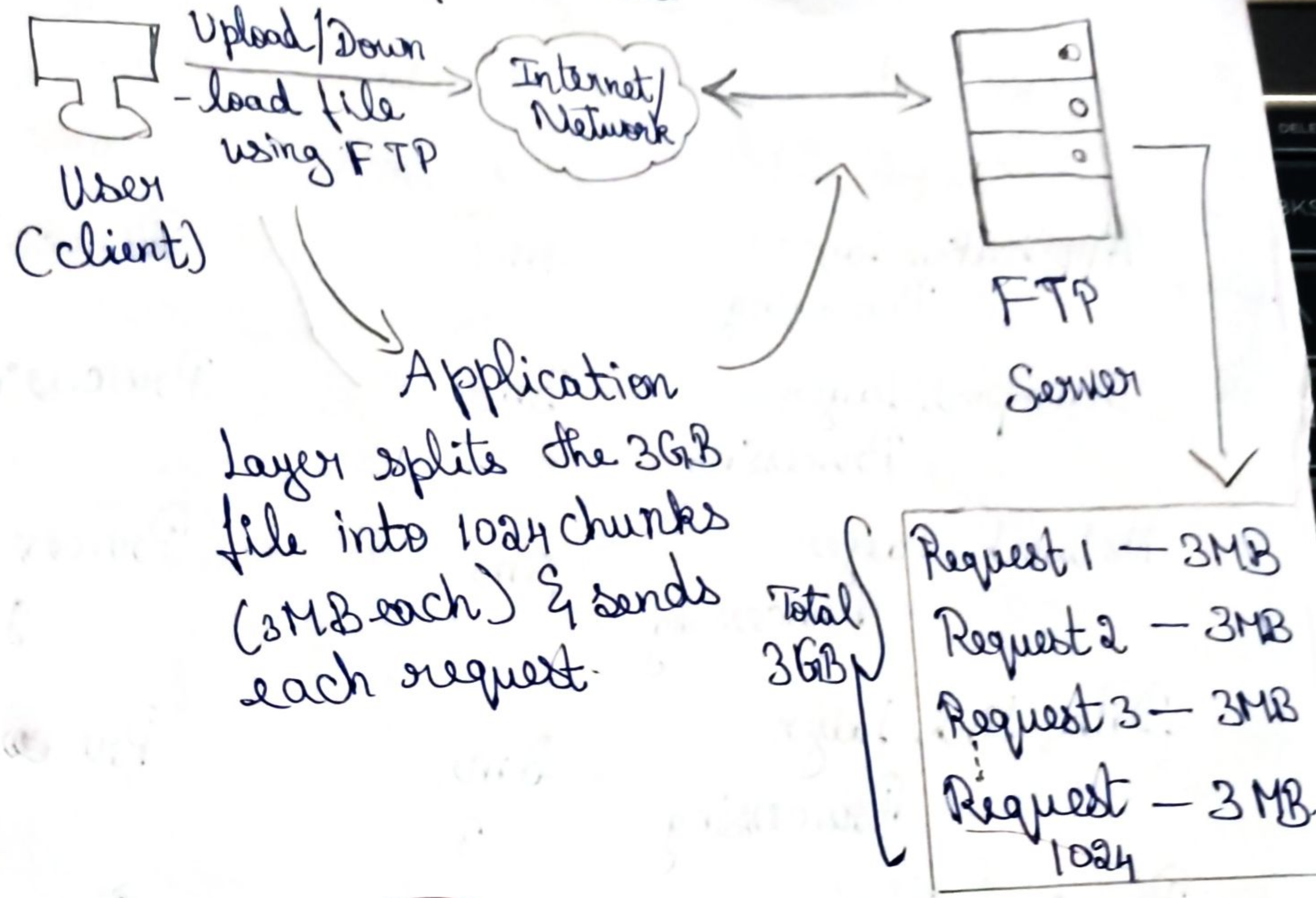
Total data = 3 GB = 3×10^9 bytes = $3 \times 10^9 \times 8$ bits

$= 24 \times 10^9$ bits.

$$\text{Transmission time} = \frac{\text{Total bits}}{\text{Throughput}} = \frac{24 \times 10^9 \text{ bits}}{120 \times 10^6 \text{ bits/sec}}$$

$= 200$ seconds.

FTP Data Transfer Process.



How the Application Layer ensures Reliable Services?

* File integrity - checks for errors and ensures file is received correctly.
* Acknowledgements - confirms successful transfer of each data block.
* Resume capability - supports restarting from the last block if the connection breaks.
* Authentication - Allows only authorized users to access the file system.
* Standardized Communication - provides a simple, user-friendly interface for file operations.

8) Soln :

1. Calculate Total End-to-End Delay.

Component	Delay (ms)	Type.
Application Layer Processing	4ms	Processing Delay
Transport Layer Processing	3ms	Processing Delay.
Network Layer Processing	5ms	Processing Delay.
Data Link Layer Processing	2ms	Processing Delay.
Physical Layer Processing	1ms	Processing Delay
Transmission Delay	12ms	Transmission Delay
Propagation Delay	18ms	Propagation Delay.

Total End-To-End Delay = 45ms.
(4+3+5+2+1+12+18).

2. How Each Layer Contributes

* Application Layer - Provides network services to user applications. It prepares data for communication.

* Transport Layer - Ensures end-to-end delivery, error recovery, segmentation, and flow control.

* Network Layer - Handles logical addressing and routing of packets across networks

* Data Link Layer - Frames the data, adds MAC addresses, performs error detection & flow control within the local network

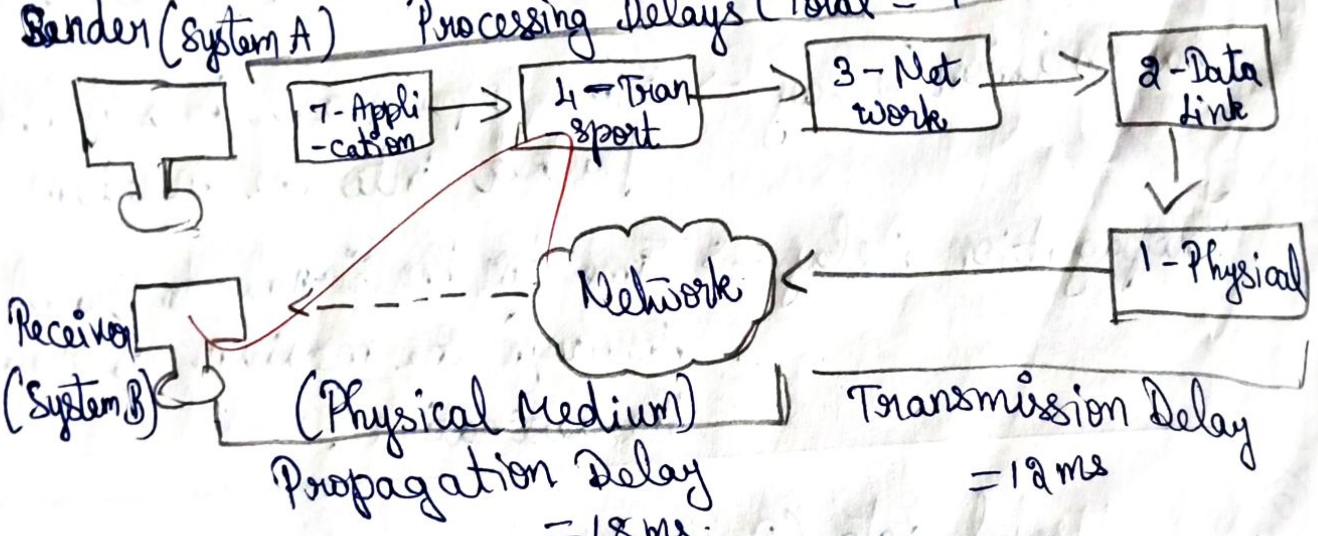
* Physical Layer - Converts bits to signals & transmits them over the physical medium.

* Transmission Delay - Time taken to push all the bits of the packet onto the physical medium.

* Propagation Delay - Time taken for the signal to travel through the physical medium from source to destination.

3. End-to-End Communication Process

Processing Delays (Total = 4+3+5+2+1 = 15ms



4. Detailed Role of Each Layer in Communication.

[7] Application Layer
* Acts as an interface between user & network
* Ex: When you open a webpage, the request is generated here.

[4] Transport Layer
* Splits data into segments, ensures reliable delivery using sequence numbers, acknowledgments
* Ex: TCP ensures the entire file is delivered

[3] Network Layer * Adds logical (IP) addresses and de[...] the best path to the destination.
* Ex: Chooses route through routers from source to destination.

[2] Data Link Layer * Converts data into frames, adds MAC addresses, detects / corrects error on local link.
* Ex: ~~cho~~ Ensures data is correctly delivered between two devices on the same network.

[1] Physical Layer * Converts frames into bits & transmits electrical / optical / radio signals over the medium.
* Ex: Sends 0s & 1s as signals through cable or wireless.

Transmission Delay → Time to push all bits of the packet into the medium

Propagation Delay → Time for the signal to travel through the medium to reach the destination.

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(9) Gn: Useful data (Payload) = 80MB

Frame size = 1600 bytes

Protocol overhead per frame = 80 bytes.

How Each Frame is Structured.

← Total Frame Size = 1600 bytes →

Useful Data 1520 bytes	Protocol over head 80 bytes

otal No of Frames Required.

↳ Useful data per frame = 1600 − 80 = 1520 bytes

No frames = $\frac{83886080}{1520}$ → Total useful data

= 55,188 frames.

2. Total Overhead Transmitted

↳ 80 × 55,188 = 4,415,040 bytes (4.21MB)

3. Transmission Efficiency

↳ Total data transmitted = 83886080 + 4415040

= 88,301,120 bytes

Efficiency = $\frac{83,886,080}{88,301,120} \times 100 = 0.94998 \times 100$

= 94.998% ⇒ $\boxed{95\%}$

4. How Protocol Overhead is Added by OSI Layers.

$\boxed{7}$ Application Layer − Application data

$\boxed{4}$ Transport Layer − TCP/UDP header

$\boxed{3}$ Network Layer − IP header

$\boxed{2}$ Data Link Layer − MAC header + FCS

$\boxed{1}$ Physical Layer − Physical framing.

$\boxed{\text{User data}}$ ↓ $\boxed{\text{+Transport header}}$ ↓ $\boxed{\text{+Network header}}$ ↓ $\boxed{\text{+Data link header}}$ ↓ $\boxed{\text{+Physical Overhead}}$

Total Overhead = 80 bytes

5. How Protocol Overhead Affects Network Performance

* Reduces efficiency
* Increases Bandwidth Usage
* Affects Throughput
* Increases Delay
* Important for Reliability & Control.

(10) Gn:

Original medical record size = 240MB

Presentation Layer compression = 30%

Transport Layer segment size = 1400 bytes

Data Link Layer Overhead = 60 bytes per segment.

Step 1: Compressed File Size.

Presentation Layer compresses 30%.

So only 70% of original data remains.

Compressed File Size = 240MB × 70%

= 168 MB

In bytes,

168 MB = 168 × 1024 × 1024

= 176,160,768 bytes

Step 2: Calculate Total no of segments.

No of segments = 176,160,768 ÷ 1400
= 125,829 segments

Step 3: Calculate Total Protocol Overhead.

Total Overhead = 125,829 × 60 = 7,549,740 bytes
= 7.20 MB.

Step 4: Calculate Final data Transmitted

Final Data = Compressed data + Protocol overhead

= 176,160,768 + 7,549,740
= 183,710,508 bytes ⇒ 175.20 MB.

How Multiple OSI Layers Work Together.

The Application Layer manges the patients medical record, while Presentation Layer Compresses & secure the data. The Transport Layer divides the data into segments & ensures reliable delivery The Network layer selects the best route to the destination. The Data Link Layer adds overhead for framing & error detection, & the Physical Layer transmits the data as signals. Together these layers ensure secure, reliable & efficient communication.